

Aircraft noise over Brockenhurst

A residents' case pack for Bournemouth Airport

Prepared by Brockenhurst residents using public flight data, 2023–2025. Not affiliated with the airport. Sources and full method are listed at the end.

In short: Brockenhurst village sits directly under the landing path for Bournemouth Airport. Aircraft are arriving **lower than the airport's own rules intend**, and often **levelling off** over the village instead of gliding down gently — and levelling off is what makes the noise. It is happening to **more flights each year, and it is worst at night**. We are not anti-airport. We would like to work with Bournemouth on sensible options: fly the quiet approach the rules already ask for, route arrivals to one side of the village over open forest, and limit night flights as the major UK airports do.

Why now — there is a live consultation. Bournemouth Airport is currently redesigning its flight paths through the formal CAA process (**Airspace Change ACP-2019-43, "FASI-South"**). The new **satellite-guided (PBN/RNAV)** routes are hyper-precise: instead of today's scattered flights, aircraft will follow the **exact same line**. If that line is drawn over Brockenhurst it becomes a **permanent, concentrated "noise highway"**. This is the moment to get the design right — and the airport has legal duties that help us: its **Section 106** planning commitment to enforce continuous descent, and its duty under the **Levelling-up and Regeneration Act 2023 (s.245)** to further the National Park's purposes (the "Sandford Principle" puts conservation first).

First, three simple ideas (no aviation knowledge needed)

- **A "quiet approach" is a gentle glide.** The quietest way for a plane to land is to come down in one smooth glide with the engines almost idling. This keeps it **higher for longer** and **quiet**.
- **"Levelling off" is the loud bit.** If a plane stops gliding and flies level for a while, it has to **push the engines back up** to hold itself up. That surge of engine power is the noise people hear over the village.
- **Where our numbers come from.** Every aircraft constantly broadcasts its own GPS position and height by radio (anyone can receive it). We used **three years** of that public data. It is the **planes' own data**, so it does not depend on anything the airport tells us.

PART 1 — The problem (and the data behind it)

1. Brockenhurst is directly under the flight path

When the wind is from the south-west (which is most of the time), planes land towards the west and approach **from over the New Forest — straight over Brockenhurst. The map below draws the actual flight path of every arriving airliner over three years, each line built from the aircraft's own GPS. They funnel into one tight stream straight over the village** and carry on to the runway.

**The path each airliner flew into Brockenhurst (runway-26 arrivals, 2023-2025)
3,083 approaches · most come straight in over the village, ~20% curve up from the south**



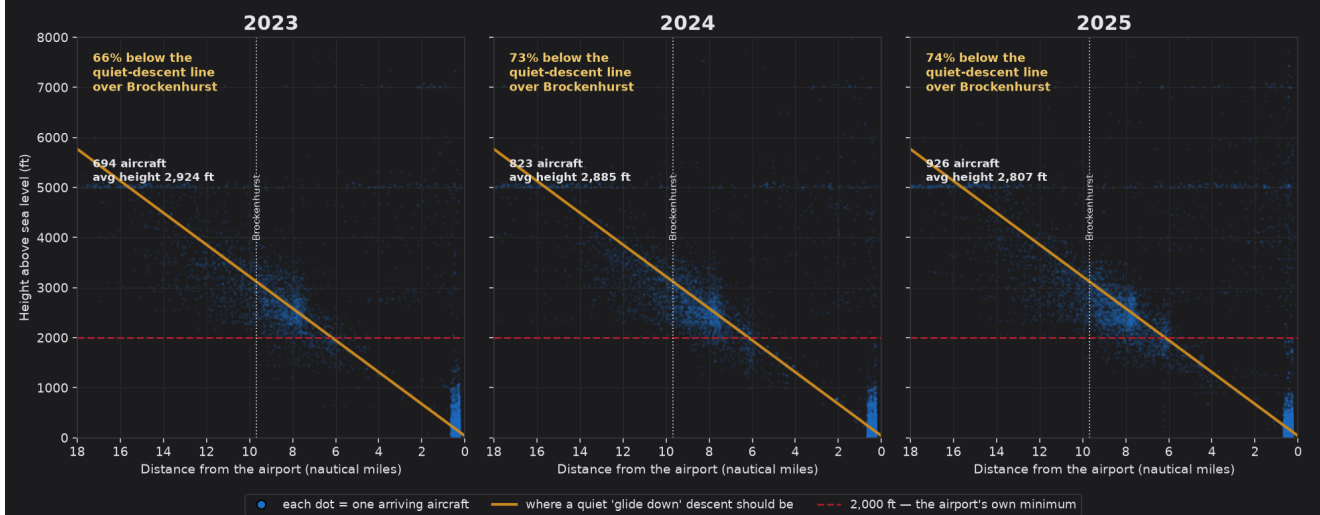
Each line is one flight through its own recorded GPS fixes (median ~9 per approach), lightly smoothed. Approximate between fixes. Large jets only, via OPDI / OpenSky.

Fact: About **76% of airliner arrivals** come in over Brockenhurst. (The rest land the other way, from the coast side, and miss the village.)

2. The planes are flying lower than they should

The orange line on the next graph is where a plane should be if it were doing the quiet glide-down — roughly **3,000 ft** as it passes Brockenhurst. Each blue dot is one aircraft. Most of them sit **below** the line: lower than a quiet descent, which means louder.

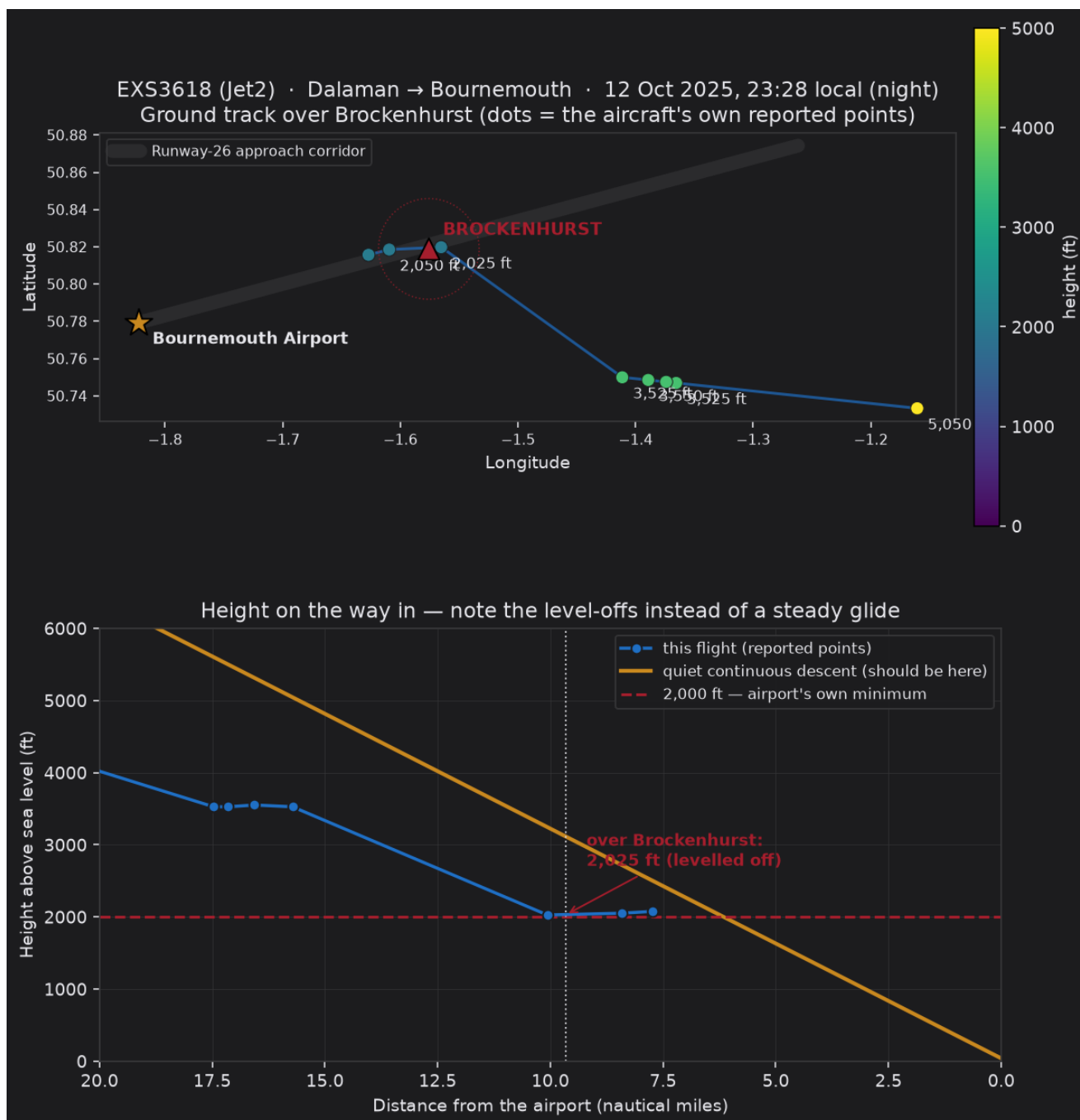
Airliners over Brockenhurst are flying below the quiet-descent line — and more so each year



Fact: The share of airliners flying **below** the quiet-descent height over Brockenhurst rose **66% → 73% → 74%** from 2023 to 2025, and the **average height is falling** (about 2,900 ft down to 2,800 ft). There are also **more of them** each year (694 → 926 with a measured height over the village).

3. They level off over the village — and that is the noise

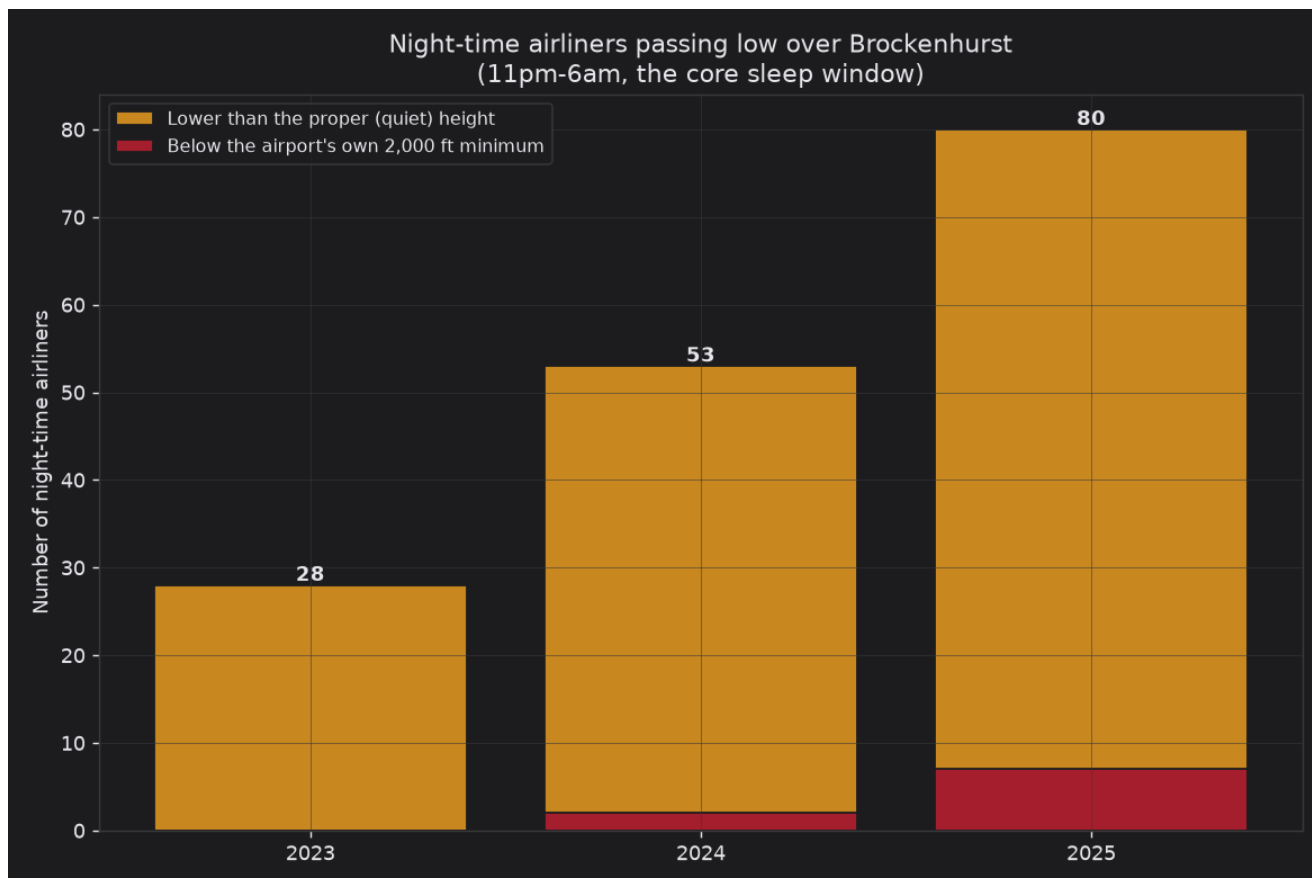
A true glide never flies level. But we can see many arrivals **stopping their** descent and flying level **right over Brockenhurst — which, as above, is exactly** when the engines have to work harder. Here is one real, recent example, built entirely from the aircraft's own recorded positions:



Real flight: A Jet2 airliner from Dalaman on the night of **12 October 2025** flew **level at about 2,050 ft directly over Brockenhurst at 23:28** — roughly 1,000 ft below a quiet descent — after also levelling off higher up. A stepped, engine-on approach, not a gentle glide.

4. It is getting worse — and worst at night

Night is when it hurts most: windows open in summer, everyone asleep. Night-time airliners over Brockenhurst have **more than doubled** in three years, and more of them are dropping below the airport's own minimum height.



Night facts: Night-time airliners over Brockenhurst rose **263 → 332 → 583** (2023 → 2025). Those coming in **below the airport's own 2,000 ft minimum** rose **0 → 2 → 7**. Aircraft **levelling off** over the village rose **63 → 95 → 157** a year.

5. This is measured against the airport's own rules

We are not inventing a standard. Bournemouth's published rules (in the official aviation manual, UK AIP EGHH AD 2.21) already say arrivals should:

- fly a "continuous descent... **at all times**,"
- stay "**as high an altitude as practical**," and
- join the approach "**without level flight**."

They even have a **night rule**: between 2130 and 0630, aircraft should get onto the final approach **no closer than 8 miles and no lower than 2,500 ft**. The data shows this is often not happening over Brockenhurst.

PART 2 — What we'd welcome working on with the airport

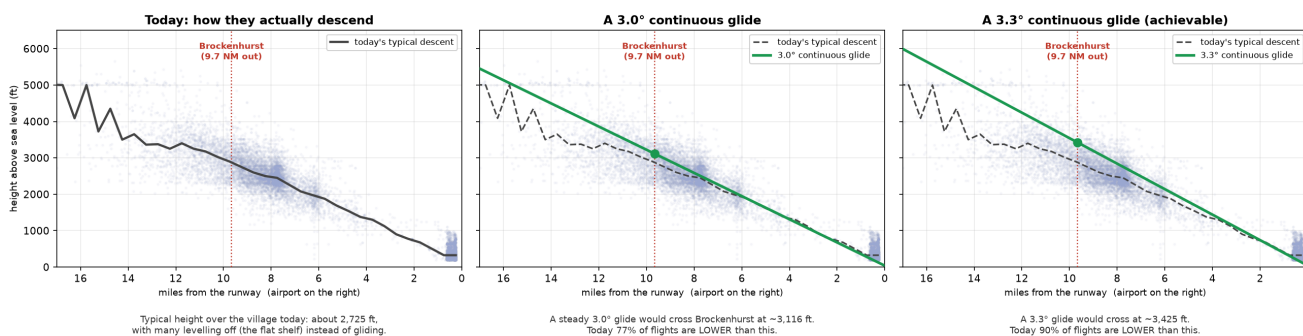
We want a constructive conversation. There are three options, and they are not mutually exclusive — the first one costs nothing and could start now.

Option A — Fly the quiet descent the rules already require

Simply **doing what the airport's own rulebook already asks for**: keep arrivals **higher and gliding down continuously**, and **stop the low level-offs**. This needs **no change to any flight path** — it is about how the descent is flown, and it uses the airport's own published policy. Higher and smoother = quieter over the village, and it could begin straight away.

The graph below shows the difference. On the left is how aircraft descend **today** — dropping low early and **levelling off** over the village (the flat shelf), typically about **2,725 ft** as they cross Brockenhurst. In the middle and right is where a steady, continuous glide would keep them: a **3.0° glide crosses** at about 3,116 ft (**today 77% of flights are lower than that**), and a slightly steeper, still-gentle **3.3° glide crosses at about 3,425 ft** (today **90%** are lower). Higher and smoother, with no level-offs, is quieter — and it is the airport's own published rule.

Option A: fly the quiet glide the rules already ask for — today vs a 3.0° and 3.3° continuous descent



Each faint dot is one recorded aircraft position (runway-26 large-jet arrivals, 2023-2025, via OPDI / OpenSky). A continuous glide keeps aircraft higher over the village and removes the level-offs that cause the noise.

Option B — Route arrivals to one side of the village (over open forest)

The **last few miles** of any landing have to line up with the runway, so those can't move. **But Brockenhurst is about 10 miles out**, and that far from the runway planes do **not** have to be on the dead-straight line. They could be guided to **join the final approach just north or south of the village — over open, unpopulated forest** — and only then turn onto the runway line. **Modern** satellite-based navigation makes this kind of "join after the village" path precise and repeatable. The green and gold dashed lines on the map in Part 1 show the idea (illustration only).

We would **not** pick north or south ourselves — each should be checked for who lives underneath. We'd ask the airport and its air-traffic partner to **assess** both **and choose the quietest option**.

Option C — Limit night flights, as the big airports already do

Britain's major airports operate strict **night-flight limits**. At **Heathrow**,

Gatwick and Stansted, **the government sets a "night quota" for 23:30–06:00: a cap on the number of night flights, a noise budget, and an outright ban on the noisiest aircraft** at night. Bournemouth is a smaller airport and isn't part of that government scheme — but it **could adopt similar measures** itself, **and many regional airports already limit night flying through their** planning agreements.

Proposed asks — we would welcome the airport:

- 1.** Publishing how well it meets its own continuous-descent rule for these arrivals, and committing to improve it (higher, smoother, no level-offs).
- 2.** Studying a new **arrival route that joins the runway line to one side of Brockenhurst**, over open forest.
- 3. Limiting and enforcing night flights** — a cap on night movements, enforcing the existing 2,500 ft / 8-mile night rule, and phasing out the noisiest aircraft at night, as the London airports do.
- 4.** Giving **Brockenhurst a seat** on the Airport Consultative Committee.

Who can make these changes happen

- **Bournemouth Airport** and its **air-traffic provider** — for how approaches are flown and designed, and for night-flight policy.
- The **Civil Aviation Authority (CAA)** — runs the formal process for any change to a flight path (an "airspace change").
- The **Airport Consultative Committee** — the airport's community forum, where residents' representatives can raise this.
- The **New Forest National Park Authority**, our **parish and district councils**, and our **MP** — natural allies, since this is about protecting a **National Park and its villages**.

How we know all this (in plain English)

- The numbers come from aircraft's **own broadcast GPS/height data** for **2023–2025**, via two well-known public datasets (OPDI, run by a European air-traffic body, and OpenSky, a research network). It is **independent of the** airport.
- **Almost every arrival is captured** (~97%). For the exact height directly over the village we rely on a smaller, representative sample, so our **counts are** minimums — **the real numbers are** higher, not lower.
- We are careful **not** to say every low flight is breaking a rule — some have air-traffic reasons. **The case is the consistent, worsening pattern across** thousands of flights, **not any single aircraft**.

Where to find the detail

Full method, definitions, sources and the flight-by-flight data are available on request (documents: Data, sources & definitions; Methodology; and an Excel

list of every measured flight). Every chart in this pack can be reproduced from public data.

Prepared from public data by Brockenhurst residents. Not affiliated with, or endorsed by, Bournemouth Airport.